INCH-POUND

MIL-P-51303B(EA)
14 November 1990
SUPERSEDING
MIL-P-51303A(MU)
9 September 1988

MILITARY SPECIFICATION

PROJECTILE, 81-MILLIMETER, SMOKE, M375A3; WP FILLING

This specification is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the white phosphorus (WP) filling of the M375A3 projectile, without fuzing components.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315

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STANDARDS

MILITARY

MIL-STD-1168 - Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

8885263 - Body Filling Assembly

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-CT, Rock Island, IL 61229-7210.)

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

90-1-378 - Interplant Pallet Assembly

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-CT, Rock Island, IL 61229-7210.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing 8885263 and subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

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3.2 Filling. The projectile, clean and free from foreign matter, shall be filled with molten WP at a temperature of 140° ± 10°F. When water is in contact with the WP during the filling operation, the water used shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. The depth of water remaining on the surface of the WP in the filled and closed projectile shall not exceed 1/4 inch. After testing, the filling shall be solidified with the projectile in a nose-up position (see 6.2 and 6.3).

3.3 Assembly. Closing, coating, and marking of the projectile shall be in accordance with Drawing 8885263.

3.4 Filling height. There shall be a 2 to 7 percent void, by total volume, existing above the surface of the water over the WP in the filled and closed projectile. The projectile shall be radiographed in a vertical, nose-up or nose-down position while the temperature of the WP is 130° ± 10°F when tested as specified in 4.2.6.2 (see 6.3).

3.5 Cleaning after closing. The exterior surface of the filled and closed projectile and the interior of the burster casing shall be thoroughly cleaned to remove any grease, oil, WP, or other foreign matter (see 6.4). No abrasive shall be used.

3.6 Leakage. The closed and cleaned projectile shall not leak when maintained at a temperature of 160° ± 5°F, for a period of 8 hours. Alternate method: the closed and cleaned projectile shall not leak when maintained at a temperature of 207° to 212°F for a minimum period of 15 minutes. The projectile shall be tested as specified in 4.2.6.1.

3.7 Workmanship. The projectile shall be free from internal and external contamination (foreign matter) and damage such as dents, cracks, and deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all require-

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ments of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.5.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.1.4 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. No more than one interfix number from one manufacturer of the following components shall be permitted in any one interfix number of filled projectiles:

(a) Burster casing
(b) Body

4.2.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.2.5 and, when specified, table I. Samples shall be selected at random.

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TABLE I. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

4.2.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. The lot represented shall be rejected when nonconformance to a critical characteristic is found. Sample items and, when applicable, the packaging thereof shall be examined and tested in accordance with the classification of characteristics in 4.2.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.2.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the

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item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.2.5 Classification of characteristics. Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table I for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.2.5(a)	Body Filling Assembly		8885263
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical				
1	No WP leakage	100% inspection	3.6	4.2.6.1
2	No WP contamination	100% inspection	3.5	VI
Major				
101	Filling not excessive	Table I, level VII*	3.4	4.2.6.2
102	No excess water	Table I, level VII*	3.2	4.2.6.2
103	Filling not insufficient	Table I, level VII*	3.4	4.2.6.2
104	Burster casing properly seated (maximum gap permitted between burster casing and body is 0.010 inch for 180°)	Table I, level VII	3.1	VI
105	Bourrelet diameter after protective finish and marking	Table I, level VII	3.1	CE
106	Protective finish evident and correct	Table I, level VII	3.1	VI
107	Marking correct	Table I, level VII	3.1	CE and VI
108	Workmanship (no damage or external contamination)	Table I, level VII	3.7	VI
Minor				
201	Marking evident and legible	Table I, level IX	3.1	VI
202	No internal contamination (to be checked prior to filling)	Table I, level IX	3.7	VI
203	No contamination after filling	Table I, level IX	3.7	VI

NOTES:

*Table I, level VII shall be used for the first three consecutive acceptable lots; thereafter, table I, level VIII shall be used.

CE – Commercial inspection equipment

VI – Visual inspection

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4.2.6 Tests.

4.2.6.1 Leakage. The projectile shall be placed in an oven in a vertical, filling-end-down position at the specified temperature and maintained for the required period of time. When testing at 207° to 212°F, the temperature of the projectile shall be measured by thermocouples or thermometers or both inserted in the burster wells of at least two projectiles in each oven batch. Leakage shall be determined by visual examination.

4.2.6.2 Filling height. The height of the WP in the filled and closed projectile shall be located radiographically while temperature of the WP is 130° ± 10°F and the projectile is oriented in a vertical nose-up or nose-down position.

5. PACKAGING

5.1 Packing, level C. The projectile shall be packed for shipment in accordance with Drawing 90-1-378-10.

5.2 Marking. In addition to any special marking required by the contract or order, the pallets shall be marked in accordance with Drawing 90-1-378-10.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

6.2 Caution. Due to the hazardous nature of WP, filling installations are advised to take all possible precautions to protect personnel from injury while filling projectiles in accordance with this specification.

6.3 Filling aid. Projectiles filled to a void of 2 to 7 percent have been found to comply with 3.4. This information is furnished as an aid to acquisition and not as a criterion for filling.

6.4 Decontamination. The filled projectiles may be cleaned by passing them through a hot water and detergent spray washer or by flushing them with hot water. Additional cleaning of the burster case may be effected with a steam lance.

6.5 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for

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evaluation and approval by the technical activity responsible for preparation of this specification.

6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

Filling height
Leakage
White phosphorus

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1315-AC80

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-51303B(EA)	PROJECTILE, 81-MILLIMETER, SMOKE, M375A3; WP FILLING

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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